Status	Finished
Started	Friday, 16 October 2026, 11:24 AM
Completed	Friday, 16 October 2026, 11:28 AM
Duration	4 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It allows for quicker identification of duplicates
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of all keys
c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are ignored
- c. Each part is searched for a specific element
- d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **5**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Backtracking
- c. Divide and conquer
- d. Greedy method

The correct answer is:
Divide and conquer

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It simplifies the sorting process by using only one pass
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Dividing the input into parts, solving each part, and combining the solutions
- c. Sorting data sequentially
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **9**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes the data unsorted
c. It complicates the selection of items
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sort()
b. order()
c. arrange()
d. sorted()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are discarded
c. They are compared element by element
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By breaking the input into smaller parts and merging them
- c. By comparing elements sequentially
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Bubble Sort
- c. Binary Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Insertion Sort
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Searching

c. Complexity

d. Merging

The correct answer is:
Complexity